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ARTICLE

# Identification of crop suitable land using geospatial techniques and assessment with socio-economic factors—case study from India

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## Abstract

Crop suitability analyses produce optimal utilisation of existing land resources for sustainable agricultural productivity. This study identified suitable locations for the main crops of West Bengal, India; rice, maize, oilseeds, pulses, wheat, barley, potatoes and vegetables using Analytical Hierarchy Process (AHP) and Geographical Information Systems (GIS). AHP is a multi-decision-criteria approach that requires parameters regarding the crop suitability analysis. Ideal parameters such as slope, elevation, rainfall, soil, land use and land cover (LULC), net sown area, river and road networks, have been selected for the specific study areas. We determined that approximately 55% of lands are under very high to moderate crop suitable zones and some districts in the north and central areas have the maximum amount of highly suitable land as compared to the other districts. The western area showed low suitability due to undulating terrain, rainfall deficiency and infertile soil. The resultant suitability data justified with secondary datasets using the Root Mean Square Error (RMSE) method showed an 11% error. Our results also correlated the socio-economic factors, agricultural productivity and agricultural labourer, with the crop suitability zones showing a positive relationship of approximately 82% and 72%, respectively. Suitability maps of the study area and their correlations with allied agricultural factors can help decision makers and planners find appropriate zones for crop production.

**Keywords** Crop suitability · AHP · GIS · Agricultural productivity · Agricultural labourer

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## 1 Introduction

Growth of population is prominent all over the world especially in the Asian countries like India, Bangladesh and China. The scope of land utilisation is being reduced for population pressure for those countries mainly rely on agriculture. According to the agricultural census in India (2015–16), total operated land has decreased to 1.11% within just 5 years (Agricultural Census 2021) and about 114.01 M ha land is now being treated as degraded and wasteland (ICAR 2010), 0.17% of Indians lag in their productive life due to starvation (Kumar and Patel 2020). Farming community need to focus more on producing major crops to meet the need of the increasing population. Identification of proper land for production and yield with maximum capability is one of the needed prerequisites for farming communities and government administrators too. A relationship between land characteristics and crop requirements is known as suitability assessment (Mustafa et al. 2011). Productivity of land for crops and water potentiality can be enumerated by land suitability evolution (FAO 2022). Suitability analysis supports to identify ideal location for crops within a specified area. Land suitability analysis must be carried out in such a way that local needs and conditions must be reflected well in the final decision. Land suitability investigation is an interdisciplinary method that assemble the information from diverse fields like soil science, meteorology, economics, management, social science. According to the Food and Agriculture Organization (FAO), current suitability and potential suitability can be measured. In this paper, present condition has been taken for the current sustainability approach.

Several criteria are needed for the evaluation (Belka 2005). Multi-criteria-decision-making (MCDM) or multiple criteria evaluation (MCE) has been advanced to progress spatial decision making when a set of substitutes needs to be evaluated based on conflicting and disproportionate criteria. MCE is an effective tool for multiple criteria decision-making process (Malczewski 2006). Geographical Information system (GIS) software helps in the application of MCDM and MCE approach in appropriate way with significant algorithms and supports to visualise the results through maps and other meaningful representations.

Application of MCDM and MCE through GIS has been proved significant and effective on spatial analysis. Different geographical areas with different features have been evaluated based on land suitability. Malczewski (2006) produced a survey literature on GIS-based multi-criteria decision analysis. Bhagat et al. (2009) conducted land suitability analysis for cereal production. They used elementary GIS technique in their research work. Pan et al. (2011) applied the concept of producing super maps for a better visualisation of land suitability; it was based on Liao city, North China. Mustafa et al. (2011) applied AHP technique to discover suitable croplands of Agra district in India, they used soil chemical and physical properties data as parameters. Samanta et al. (2011) implied MCDM techniques in Morobe province in Papua New Guinea for refined better place for rice cultivation. They selected five parameters: topography, climate, land accessibility, physical and chemical properties of soil. Yalew et al. (2016) identified suitable agricultural land in the upper Blue Nile basin in Ethiopia using MCDM techniques. They incorporated topography, soil

properties, climate, land use, road network as parameters of the study. Bozdağ et al. (2016) applied this method for Cihanbeyli in Turkey to determine suitable agricultural land. The evaluating criteria of their study were topography, soil, climate and groundwater. Ahmed et al. (2016) applied GIS-based agricultural land suitability to determine suitable agricultural land for rubber crops in Malaysia. They considered topography, climate, rainfall, soil properties as parameters. Kazemi and Akinci (2018) applied MCDM and GIS systems in Golestan Province, Iran to identify rainfed farming zones. They used seasonal rainfall data as major parameter in the study. An extensive literature review about crop suitability analysis showed that the type of parameters varies with varying geographical locations (Table 1).

Geographically, West Bengal is diversified and economically depends on agricultural activities and shares a significant proportion from the total agricultural production of India. The aim of the study focuses on the application of AHP techniques to determine the suitable crop lands of West Bengal. The main objectives of the study are (i) to identify the crop suitable zones and (ii) correlation between suitability and allied agricultural factors such as agricultural labourer and agricultural production. Several studies (Santra and Mitra 2016; Rukshana and Molla 2021; Sarkar et al 2021) have already covered small portion of the state on related subject matter but in this paper, the whole state has been selected as study area. Novelty of this article lies in application of crop suitability analysis for physically and economically vast diversified heterogeneous region where it is generally used within a comparatively smaller homogeneous boundary. According to the authors' knowledge, this kind of MCDM technique already applied in West Bengal, India within a specific geographical boundary, for the first-time authors of this paper attempted to analyse whole state. Also, authors attempted to correlate their observation with real-world socio-economic factors of the study area to understand actual utilisation of crop suitable land from economic and social perspective also. Validation process that has been applied in this paper is also quite unique compared with other contemporary research article related to this field.

## 2 Study area

West Bengal is the 13th largest Indian state by area. It lies between 85°50′E and 89°50′E and 21°25′N and 27°13′N. Geographical diversity is prominent in West Bengal. Mainly the state is divided into four major geographic divisions: Himalaya Mountain Region in the north, Chota Nagpur Plateau in the west, Lower Gangetic Plain and coastal belts on the south. There are three watershed basin like Lower Ganga Basin, Damodar Basin and Teesta Basin maintaining affluent water resources through Ganga, Damodar and Teesta. The entire state is under high rainfall region compared to other states (WRIS 2022). The climate here varies from moist tropical to dry tropical and from sub-tropical to temperate. Recently, West Bengal is divided into 23 districts as few large districts has been subdivided for administrative purpose. In this paper, previous major 19 districts have been utilised for the analysis as newly formed districts were part of it. This state holds leading position in rice